

ICS Security Analysis Report

Process-Aware Analysis of SA08 and MA06 Attack Scenarios

Based on the HAI Industrial Control System Dataset

1. Executive Summary

Industrial Control System (ICS) security analysis fundamentally differs from traditional IT security monitoring due to the tight coupling between cyber actions and physical process behavior. This project presents a **process-aware security analysis** of two industrial attack scenarios—**SA08** and **MA06**—from the HAI ICS dataset.

The analysis focuses on understanding **normal control behavior** before assessing potential attack impact, using **time-series visualization of setpoints (SP), process variables (PV), and control variables (CV)**. Rather than assuming that attacks must immediately produce visible anomalies, this project deliberately evaluates whether the available evidence is sufficient to support a security conclusion.

The key outcome of this work is the demonstration that **early-stage or stealthy industrial attacks may not produce clear, visually detectable deviations**, emphasizing the need for cautious interpretation, contextual investigation, and multi-layered detection strategies in real-world ICS environments.

2. Dataset and System Context

2.1 HAI ICS Dataset Overview

The HAI ICS dataset represents a simulated industrial environment consisting of multiple interconnected process units, including pressure, level, flow, and temperature control loops. The dataset provides both **normal operation data** and **attack scenarios**, recorded as time-series sensor and actuator signals.

Each attack scenario is associated with a high-level description of the attacker's intent, such as manipulation of setpoints, replay of sensor values, or modification of control parameters.

2.2 Control System Model: P1-LC (Level Control)

This project focuses on the **P1-LC (Level Control)** subsystem, which regulates the water level of a return water tank. The control loop consists of:

- **Setpoint (SP)**
Operator-defined target water level (e.g., P1.B3004)
- **Process Variable (PV)**
Measured water level from the level sensor (e.g., P1.LIT01)
- **Control Variable (CV)**
Actuator command to the level control valve (e.g., P1.LCV01D)

Under normal operation, the controller adjusts the control valve to minimize the deviation between SP and PV while maintaining stable process dynamics.

2.3 Security-Relevant Variables

This analysis adopts a **minimal-variable strategy** to reduce noise and maintain interpretability:

- SP → operator or attacker intent
- PV → physical process response
- CV → controller reaction

This choice reflects realistic industrial incident analysis, where excessive variable inclusion can obscure causal relationships.

3. Methodology

3.1 Analysis Philosophy

The analysis follows a **process-aware, evidence-driven methodology**:

1. **Understand normal behavior first**
Establish the expected relationship between SP, PV, and CV during stable operation.
2. **Compare against attack data without presupposition**
Avoid assuming that the presence of an attack label implies observable anomalies.
3. **Refuse unsupported conclusions**
If available evidence is insufficient, explicitly state uncertainty.

This approach aligns with real-world ICS incident response, where premature conclusions can lead to incorrect root-cause attribution.

3.2 Scenario Selection Rationale

SA08 was selected as the initial analysis target because it is the **first attack in the dataset timeline**, minimizing potential contamination from prior attack effects.

MA06 was analyzed subsequently to provide contrast and evaluate whether different attack mechanisms result in more observable process deviations.

3.3 Visualization-Based Analysis

Time-series plots of SP, PV, and CV were generated for both **normal** and **attack** datasets. Visual inspection focused on:

- SP–PV tracking behavior
- Control stability and oscillations
- CV response magnitude and timing

This project intentionally avoids machine-learning-based anomaly detection to emphasize **engineering interpretability** over automated classification.

3.4 Visualization-Based Analysis

To ensure consistency and reproducibility, fixed time windows were selected for both normal operation and attack conditions.

The selected intervals reflect stable process behavior for baseline comparison and the documented execution period of each attack scenario.

SA08 – Time Windows

- **Normal operation**
 - Start: 2019-11-01 20:00:00+09:00
 - End: 2019-11-01 21:00:00+09:00
- **Attack scenario**
 - Start: 2019-10-29 13:40:00+09:00
 - End: 2019-10-29 14:40:00+09:00

MA06 – Time Windows

- **Normal operation**
 - Start: 2019-11-01 20:40:00+09:00
 - End: 2019-11-01 21:00:00+09:00
 - **Attack scenario**
 - Start: 2019-11-01 09:20:00+09:00
 - End: 2019-11-01 09:40:00+09:00
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4. SA08 Attack Scenario Analysis

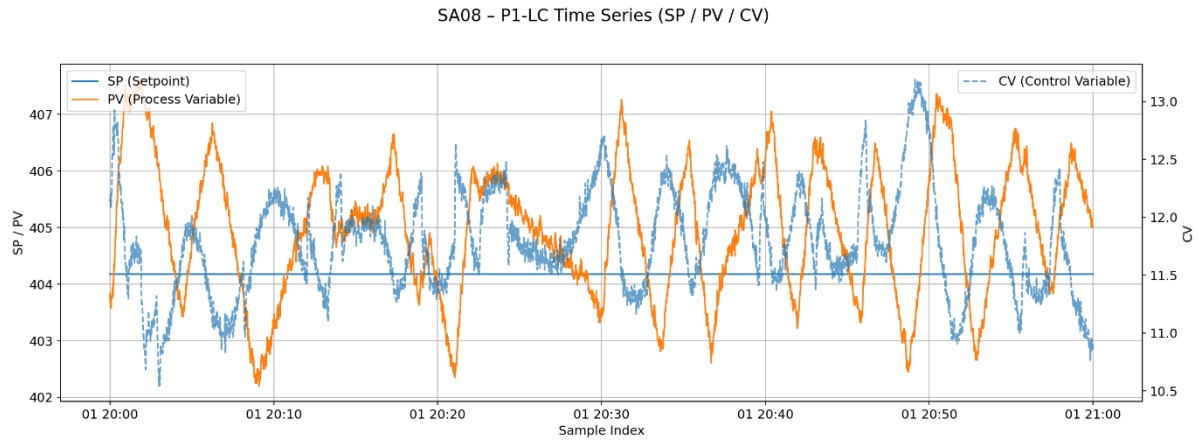
4.1 Attack Description (SA08)

SA08 targets the P1-LC control loop by **manipulating the setpoint** of the level controller. According to the scenario description, the attacker increases the SP and subsequently restores it using a trapezoidal profile, with the intent of hiding the manipulation from the human-machine interface (HMI). Additionally, the attack involves replaying previously recorded sensor values.

4.2 Normal Operation Behavior

Under normal conditions, the following behavior is observed:

- PV closely tracks SP with minimal steady-state error
- CV adjusts smoothly to compensate for disturbances
- No sustained oscillations or instability are present

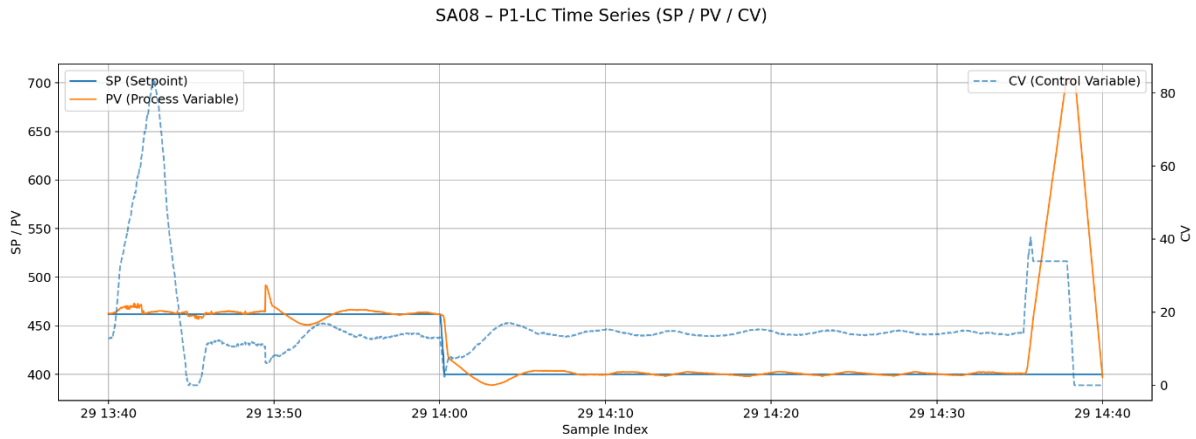


【Figure 1: SA08 – Normal Operation Time-Series (SP / PV / CV)】

This behavior establishes a baseline for evaluating potential attack effects.

4.3 Attack Data Observation

In the attack dataset, the SP, PV, and CV signals were plotted using the same scale and time resolution as the normal dataset.



【Figure 2: SA08 – Attack Time-Series (SP / PV / CV)】

Key observations:

- No abrupt or sustained deviation between SP and PV is visible
- CV behavior remains within expected operational bounds
- Overall control loop dynamics appear consistent with normal operation

4.4 Assessment

Based solely on the available time-series visualization:

No conclusive evidence of abnormal process behavior can be derived for SA08 at this stage.

This does **not** imply that the attack was ineffective. Instead, it indicates that:

- The attack may be intentionally stealthy
- The manipulated signals may fall within acceptable control tolerances
- Additional data (e.g., controller internals, network traffic, higher-resolution timing) would be required for confirmation

This outcome reflects a realistic industrial security challenge: **the absence of obvious anomalies does not equate to the absence of malicious activity.**

5. MA06 Attack Scenario Analysis(SA08 + SA14)

5.1 Attack Structure and Context

MA06 is a **composed attack scenario**, consisting of two individual attack patterns:

- **SA08** – Setpoint manipulation combined with sensor replay
- **SA14** – Additional malicious manipulation affecting the same control loop

As a result, MA06 must be analyzed as a **multi-stage attack**, rather than a single, isolated event.

Each component attack is evaluated under both **normal** and **attack** time windows, leading to four representative visualizations.

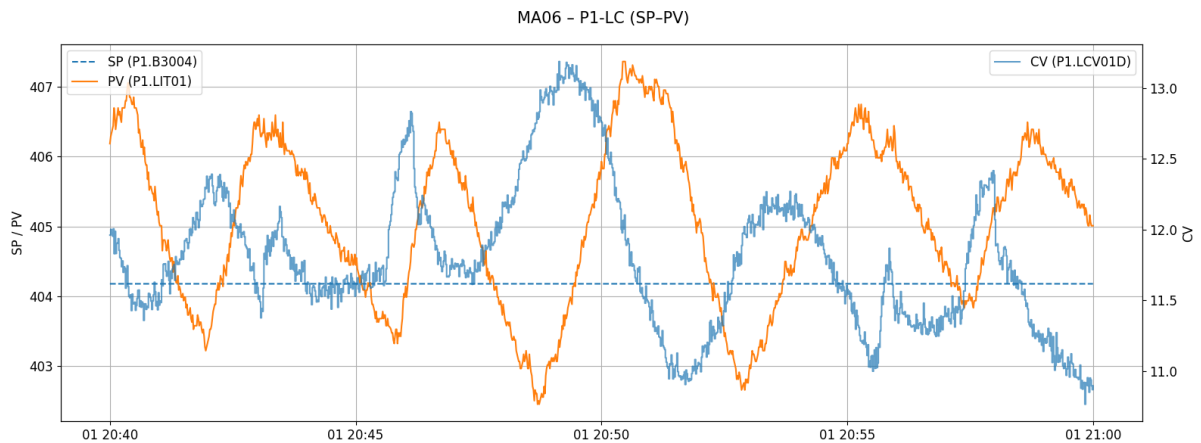
5.2 Visualization Scope

The following datasets were analyzed as part of MA06:

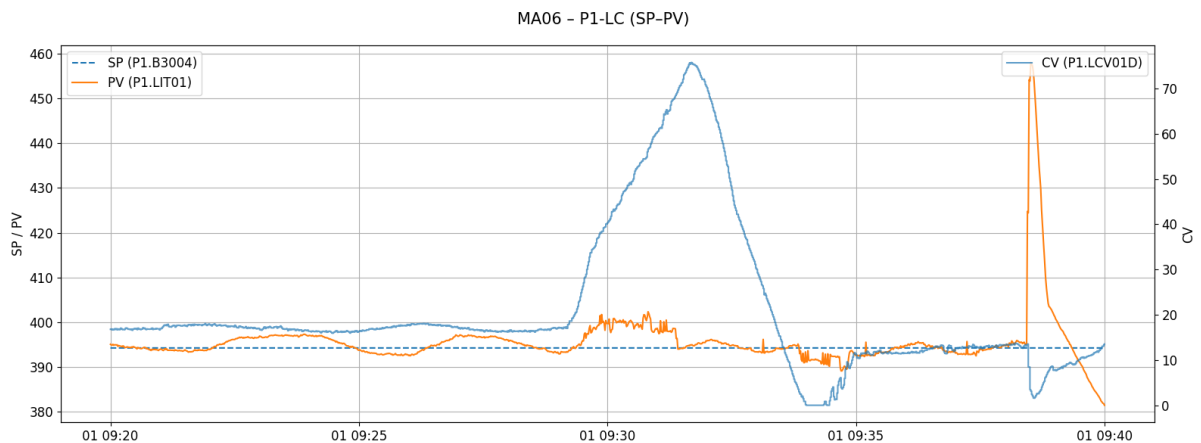
- SA08 – Normal operation
- SA08 – Attack window
- SA14 – Normal operation
- SA14 – Attack window

This structure reflects the **composed nature of MA06**, not data instability or visualization redundancy.

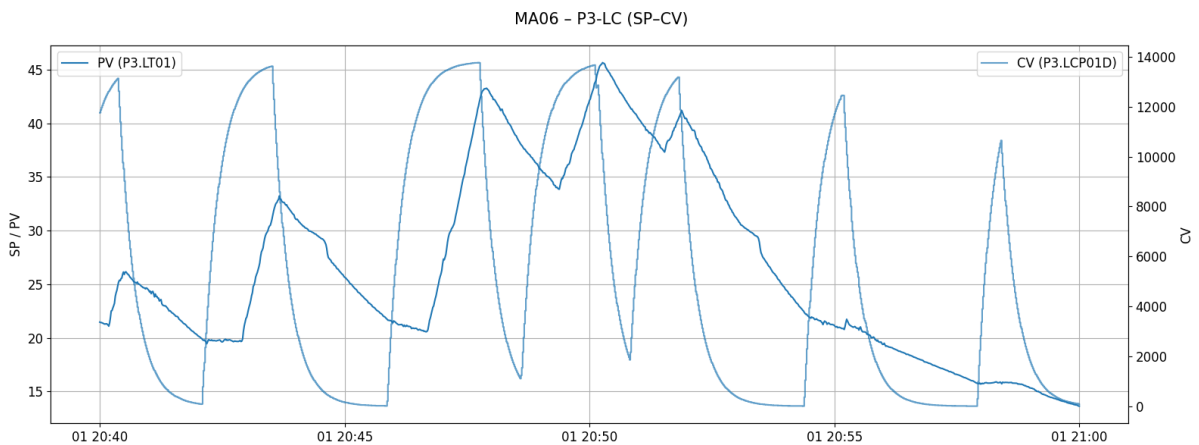
Only representative figures are included in the main report.
Additional plots are available in the project repository.



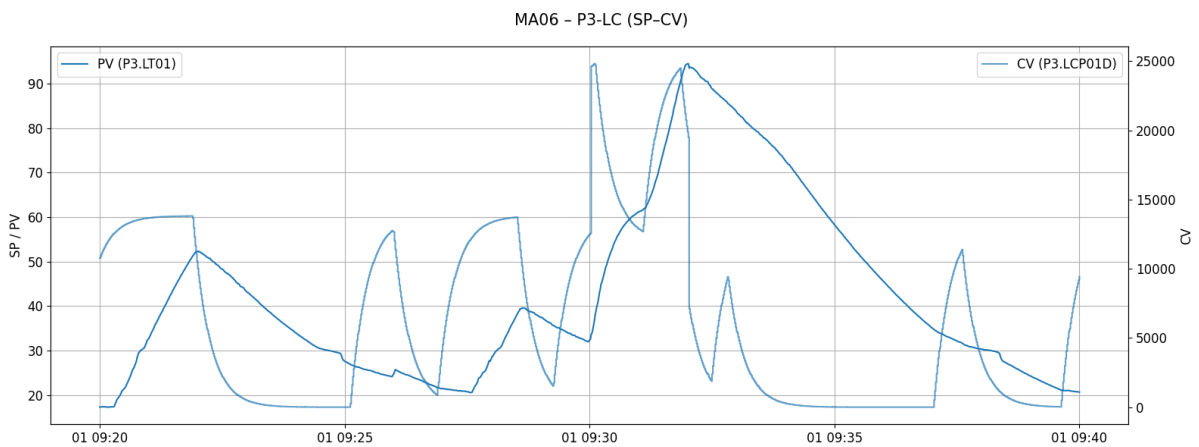
【Figure 3: MA06–SA08 – Normal Operation Time-Series (SP / PV / CV)】



【Figure 4: MA06–SA08 – Attack Time-Series (SP / PV / CV)】



【Figure 5: MA06–SA14 – Normal Operation Time-Series (SP / PV / CV)】



【Figure 6: MA06–SA14 – Attack Time-Series (SP / PV / CV)】

5.3 Observations

Across both SA08 and SA14 components:

- SP, PV, and CV relationships remain largely within expected operational bounds
- No single visualization provides definitive evidence of process disruption
- Observable behavior remains plausible from a pure control-system perspective

Importantly, the absence of a clear anomaly must be interpreted in the context of a **composed attack strategy**, where each individual action may be designed to remain subtle.

5.4 Assessment

The MA06 scenario demonstrates a critical challenge in ICS security analysis:

Composed attacks may distribute their impact across multiple stages, such that no individual stage produces a clearly detectable anomaly.

This reinforces the limitation of relying solely on surface-level process variables and highlights the need for:

- Cross-stage correlation
 - Context-aware detection
 - Controller- or network-level inspection
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6. Security Interpretation

6.1 Implications for Detection Strategies

The findings highlight the limitations of:

- Simple threshold-based alarms
- Single-variable anomaly detection
- Visual inspection without process context

In particular, SA08 demonstrates that **rule-based detection may fail during early or carefully crafted attack phases.**

6.2 Relevance to Industrial SOC Operations

From a SOC or OT security perspective, this project illustrates that:

- Alerts must be contextualized with process knowledge
 - “No alert” does not equal “no incident”
 - Escalation decisions often require additional investigative layers
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7. Limitations

This project intentionally operates within defined boundaries:

- Only time-series process data was analyzed
- No network traffic or PLC logic was available
- No precise ground-truth timestamps for attack initiation were used
- No machine learning or statistical anomaly detection was applied

These limitations are representative of **early-stage or resource-constrained investigations** in operational environments.

8. Future Work

Potential extensions of this project include:

- Correlating process data with network traffic (e.g., Modbus write operations)
 - Incorporating controller logic or parameter inspection
 - Evaluating detection performance of rule-based vs. model-based approaches
 - Applying the same methodology to additional HAI attack scenarios
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9. Conclusion

This project demonstrates a disciplined, process-aware approach to ICS security analysis that prioritizes **evidence over assumption**. By analyzing SA08 and MA06 without forcing conclusions unsupported by observable data, the work reflects the realities of industrial incident response.

The key takeaway is not the identification of a specific attack signature, but the recognition that **industrial attacks can remain operationally invisible**, particularly during early or stealthy phases. Effective ICS security therefore requires cautious interpretation, layered monitoring, and deep process understanding.